



COURSE SLIDES

Generative AI for Project Management

Course Code: C013

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Course Trainer Profiles

1

Assigned Tertiary Infotech Academy Trainer

Project-management and generative-AI facilitator who guides evidence-led planning, delivery, control and hands-on application to the synthetic BeaconWorks scenario.

Let's Know Each Other

1 Your name and organisation / role.

2 Your experience with this subject (if any).

3 What you want to be able to do after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

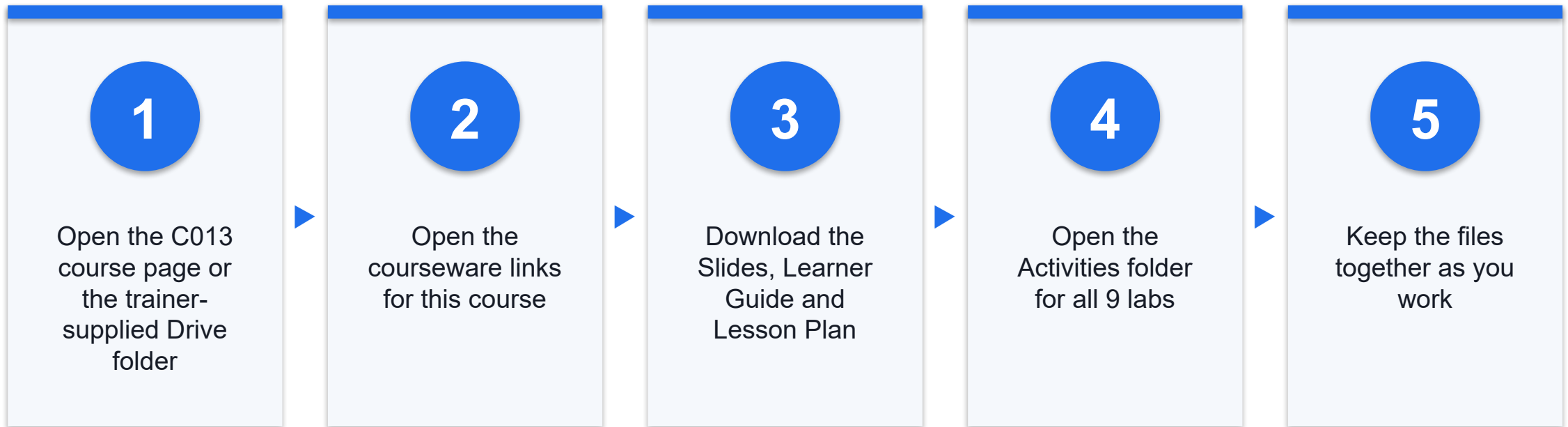
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Access Course Materials



Course page: <https://www.tertiarycourses.com.sg/effective-project-management-with-generative-ai-genai.html>

Lesson Plan — 2 Days, 15 instructional hours

Course sequence

- Day 1 — Initiate and Plan — establish evidence, authority and an integrated baseline
 - Topic 1: Project Initiation and Planning with GenAI (Labs 1–3)

Day 2

- Day 2 — Execute, Control and Close — deliver with traceable decisions and responsive controls
 - Topic 2: Project Execution and Delivery with GenAI (Labs 4–6)
 - Topic 3: Manage Project Risks with GenAI (Labs 7–9)
- Daily timing
 - 9:30 am–6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)

Learning Outcomes

1

Initiate & Plan

Develop project parameters and an integrated project plan with GenAI according to business requirements.

2

Execute & Deliver

Resolve project delivery challenges by using GenAI to support people, knowledge, quality, communications, resources and budgets.

3

Respond & Change

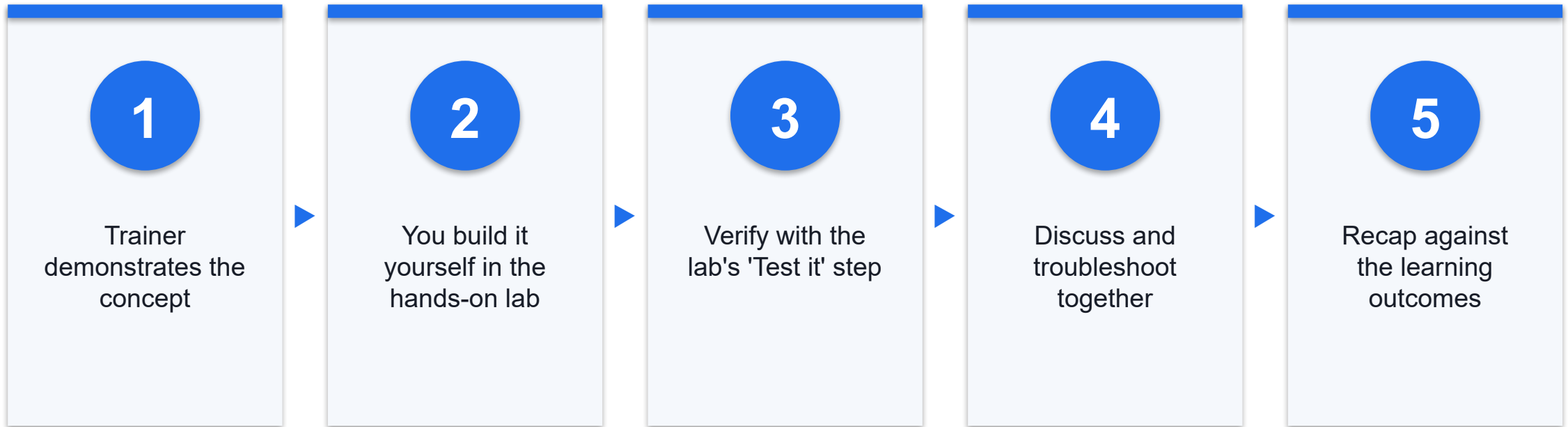
Anticipate, monitor and respond to project risks and changes with GenAI while keeping decisions evidence-led and human-owned.

4

Control & Close

Control project performance, obtain acceptance and close a project or phase with traceable records and reusable lessons.

How You'll Learn





CORE CONCEPTS

Course Fundamentals

The Human-Owned GenAI Project System

1

Ground

Use approved project evidence and make the source boundary explicit.

2

Generate

Ask for options, structures, critiques and transformations—not hidden authority.

3

Check

Verify facts, calculations, commitments, confidentiality and uncertainty.

4

Decide

Name the human owner, rationale, approval and update to the project record.

One Integrated Project Loop

1

Initiate

Authorise purpose, boundaries, stakeholders and decision rights.

2

Plan

Integrate scope, schedule, cost, quality, resources, communications, risk and procurement.

3

Execute

Authorise work, manage people and knowledge, produce and verify deliverables.

4

Control

Compare evidence with baselines, respond to risk and decide changes.

5

Close

Accept, transition, archive and convert experience into reusable lessons.

COURSE OPERATING PRINCIPLE

AI accelerates project thinking; people own project truth.

Every generated output is a draft until its evidence, assumptions, calculations and authority are checked.

What You Will Build

1

Initiation Pack

AI working agreement · charter, assumptions and source ledger

2

Integrated Plan

stakeholder and communications plan · scope, schedule, cost, quality, resource, risk and procurement baseline

3

Delivery System

work package and decision records · quality, team and status evidence

4

Control & Closure

risk and change decisions · acceptance, transition and lessons pack

How Every Lab Progresses

1

Open the verified synthetic project checkpoint from the prior lab.

2

Run a bounded prompt against supplied evidence and preserve unknowns.

3

Apply project-management logic, formulas and decision criteria.

4

Perform a named human review and record the final decision.

5

Test the artifact against observable checks and save the next checkpoint.

TOPIC 01

Project Initiation and Planning with GenAI

AI working method · charter and stakeholders · integrated management plan · scope, schedule, cost, quality, resources, communications, risk and procurement

Key Concepts — Project Initiation and Planning with GenAI

1

Human-owned AI workflow

Use GenAI to propose, structure and challenge; a named project owner validates every decision.

2

Project charter

Connect the business need, measurable objectives, boundaries, authority and success conditions.

3

Stakeholder strategy

Separate influence, interest, impact and information needs before choosing an engagement approach.

4

Integrated baseline

Link deliverables, work packages, dependencies, resources, costs and acceptance evidence.

5

Management plans

Define how scope, schedule, cost, quality, communications, risks and suppliers will be managed.

6

Planning uncertainty

Record assumptions, confidence, sources and review triggers instead of hiding uncertainty in polished prose.

Use GenAI as a Project Co-Pilot

WHAT

- A generative AI assistant creates likely text, tables and alternatives from the instructions and context it receives. For project work it is best treated as a co-pilot: useful for organising supplied evidence, generating options and challenging a draft, but not an accountable decision-maker or a source of project truth.

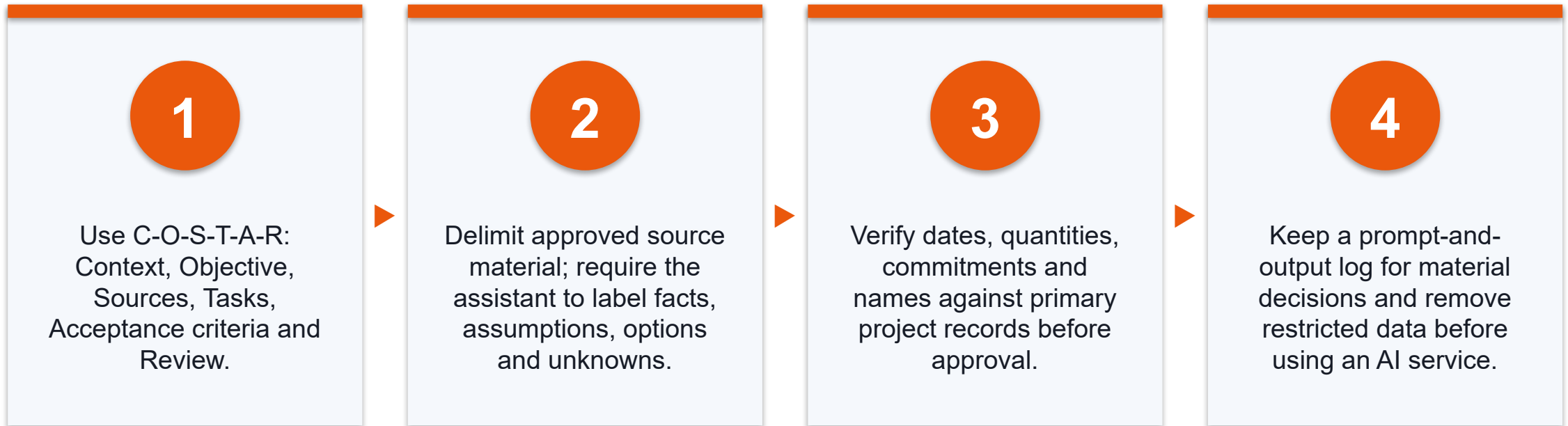
WHY IT MATTERS

- Project records affect money, commitments, people and suppliers. Fluency can disguise a missing source, an invented dependency or an unrealistic estimate. A controlled workflow makes the human owner, evidence boundary and review gate visible before an output enters the project record.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Use GenAI as a Project Co-Pilot



In Practice — Use GenAI as a Project Co-Pilot

WORKED EXAMPLE

- The project manager supplies the approved project brief and asks for a charter draft, not a new business case.
- The response includes a source ledger and marks an unstated target date as UNKNOWN.
- The sponsor selects the objective and the project manager records the final wording and rationale.

DECISION GUIDE

- **Use when**
 - The task is to structure, summarise, compare, draft, critique or explore options from approved inputs.
 - A human owner can check the result against evidence and has authority to approve the final record.
- **Avoid when**
 - The prompt would expose confidential, personal or security-sensitive information to an unapproved service.
 - The output would create a commitment, forecast or decision without evidence and named human approval.

Initiate the Project and Develop the Charter

WHAT

- Initiation turns a business need into an authorised project. A practical charter names the purpose, measurable objectives, high-level deliverables, exclusions, constraints, major assumptions, milestone horizon, initial risks, sponsor and project manager authority. It is a decision record, not a detailed plan.

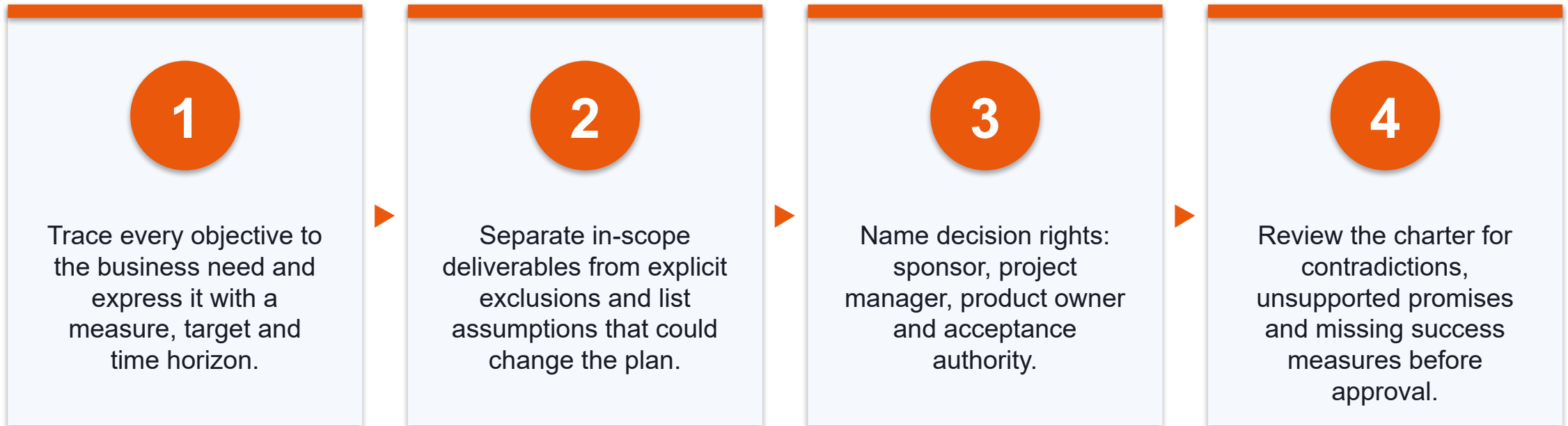
WHY IT MATTERS

- Teams that begin with solution activity before confirming the problem and authority often optimise the wrong work. GenAI can expose ambiguity and draft alternatives, but only the sponsor and accountable leaders can decide which outcome, tolerance and boundary the organisation accepts.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Initiate the Project and Develop the Charter



In Practice — Initiate the Project and Develop the Charter

WORKED EXAMPLE

- **Need:** reduce customer-service response delays during seasonal peaks.
- **Objective:** release an approved self-service portal pilot by week 12 within SGD 120,000, with named acceptance checks.
- **Exclusion:** replacement of the existing case-management platform; this prevents silent expansion.

DECISION GUIDE

- **Use when**
 - A new project or phase needs shared authority, boundaries and measurable success conditions.
 - Several stakeholders interpret the business need differently and require one approved reference point.
- **Avoid when**
 - Using a charter as a substitute for detailed requirements, estimates or a delivery plan.
 - Allowing AI-generated objectives or dates to become commitments before sponsor review.

Identify Stakeholders and Plan Engagement

WHAT

- A stakeholder is a person, group or organisation that can affect, be affected by or perceive itself to be affected by the project. Stakeholder planning combines identification, influence and interest analysis, impact, current and desired engagement, information needs, decision rights and an owner for each engagement action.

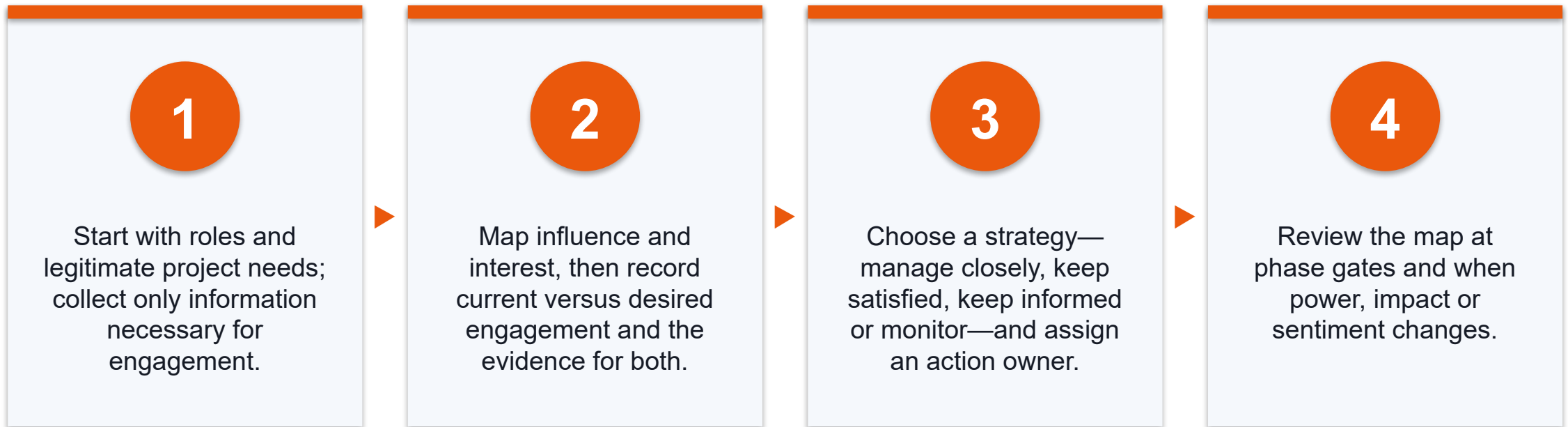
WHY IT MATTERS

- A generic mailing list treats every stakeholder as if the same information and cadence were useful. Analysis helps the team focus attention where decisions, adoption, resistance or operational impact matter, while avoiding unnecessary personal data.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Identify Stakeholders and Plan Engagement



In Practice — Identify Stakeholders and Plan Engagement

WORKED EXAMPLE

- The service director has high influence and owns acceptance; the support agents have high impact and essential workflow knowledge.
- The plan gives the director a fortnightly decision brief and the agents weekly design reviews with visible action tracking.
- AI suggests message variants from role-based needs; the project manager checks tone and factual accuracy.

DECISION GUIDE

- **Use when**
 - Planning approvals, requirements discovery, change adoption, communications and escalation routes.
 - A project affects multiple functions with different concerns or decision rights.
- **Avoid when**
 - Inferring sensitive traits, private motivations or sentiment without evidence.
 - Treating a power-interest grid as permanent or as a substitute for direct conversation.

Plan Scope with Requirements, Deliverables and a WBS

WHAT

- Scope planning converts needs into agreed requirements, deliverables, acceptance criteria and a work breakdown structure. A WBS decomposes the total project scope into manageable work packages; its dictionary explains each package, owner, boundaries and completion evidence. Decomposition describes outcomes before activities.

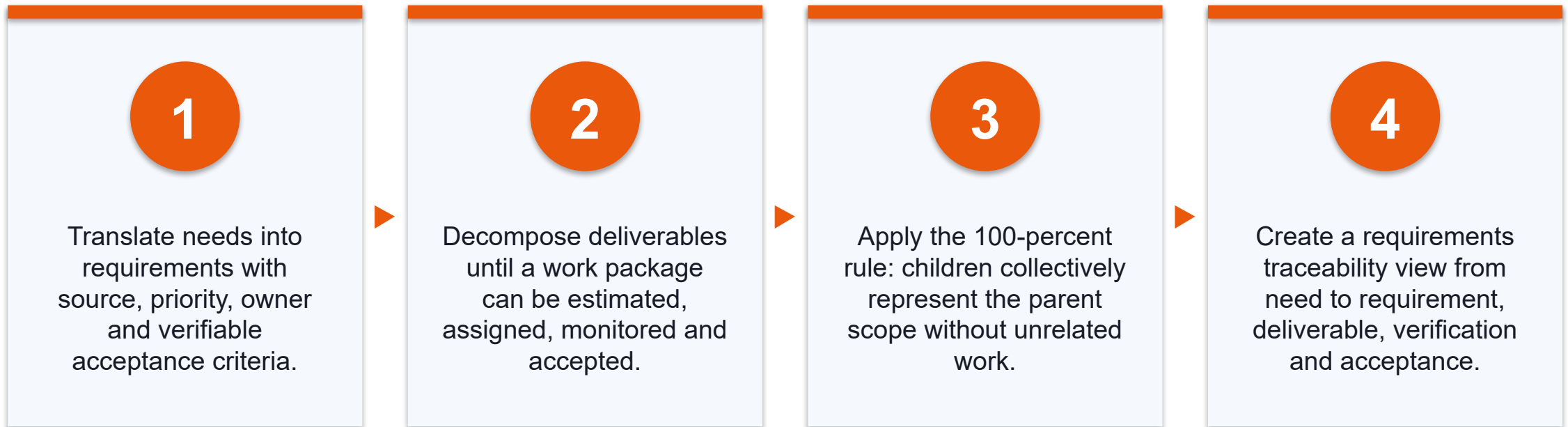
WHY IT MATTERS

- A schedule or budget built from vague nouns is fragile. The WBS creates a common reference for scope, schedule, cost, risk and responsibility, while acceptance criteria make completion observable. GenAI can find gaps and propose decomposition, but the team must confirm that the WBS covers exactly the approved scope.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Plan Scope with Requirements, Deliverables and a WBS



In Practice — Plan Scope with Requirements, Deliverables and a WBS

WORKED EXAMPLE

- Deliverable 1.0 Portal Pilot decomposes into 1.1 Content, 1.2 Configuration, 1.3 Integration, 1.4 Testing and 1.5 Launch readiness.
- Work package 1.4 has an owner, test environment, entry criteria, exit evidence and explicit exclusions.
- AI critiques the hierarchy for overlap; the team resolves the final boundary.

DECISION GUIDE

- **Use when**
 - Building estimates, assigning ownership, controlling changes or defining acceptance.
 - Requirements need traceability across business, technical and operational work.
- **Avoid when**
 - Creating a task list with no deliverable hierarchy or acceptance evidence.
 - Accepting AI-created requirements that cannot be traced to an authorised stakeholder or source.

Build a Credible Schedule and Cost Baseline

WHAT

- A credible schedule sequences the complete work, uses realistic durations, links dependencies, identifies resources and shows the critical path and margin. A cost baseline time-phases approved estimates so planned value can later be compared with earned value and actual cost. Both baselines depend on the same WBS.

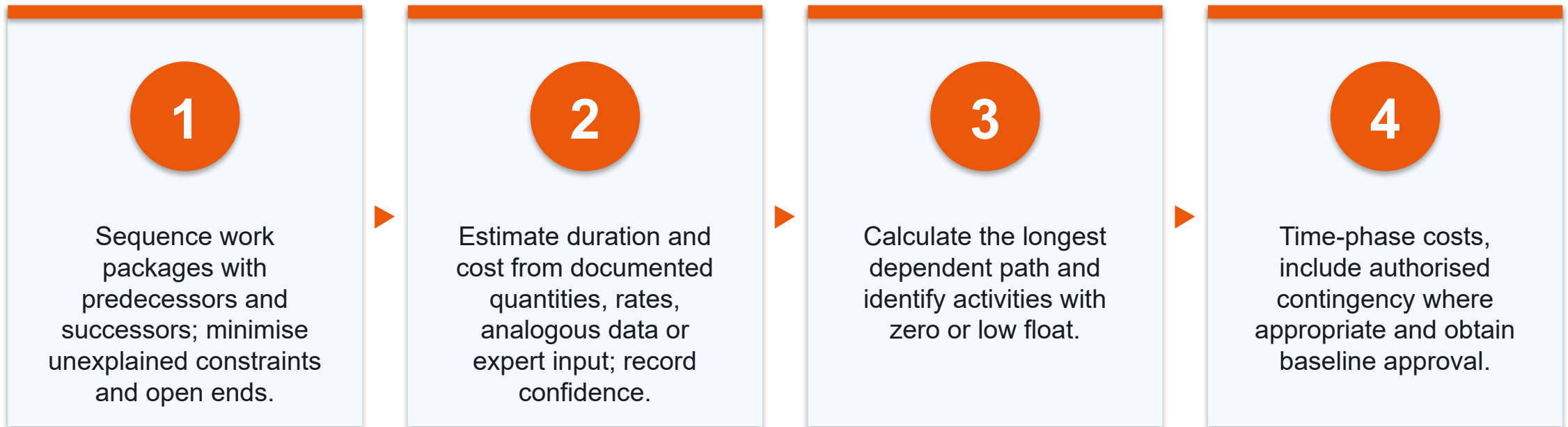
WHY IT MATTERS

- A list of dates does not show whether the finish is achievable. Network logic reveals which delay matters, while documented estimate assumptions make uncertainty discussable. GenAI may help normalise data or explore scenarios, but it must not invent durations, rates or dependencies.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Build a Credible Schedule and Cost Baseline



In Practice — Build a Credible Schedule and Cost Baseline

WORKED EXAMPLE

- Content approval precedes portal configuration; integration test follows both configuration and test-data readiness.
- The critical path runs through content approval, configuration, integration test and acceptance.
- A two-week content delay moves the finish unless the team changes logic, scope or capacity through an approved decision.

DECISION GUIDE

- **Use when**
 - Testing whether a target date and budget are realistic and creating a performance reference.
 - Analysing the consequence of delay, resource limits or an approved change.
- **Avoid when**
 - Using fixed target dates as if they were estimates or hiding uncertainty in a single precise number.
 - Optimising a schedule generated by AI when its logic and source data have not been checked.

Integrate Quality, Resources, Communications, Risk and Procurement Plans

WHAT

- The project management plan is an integrated set of baselines and management approaches. Quality planning defines standards, metrics and verification; resource planning defines roles, capacity and responsibility; communications planning matches message to audience and decision; risk planning defines categories, thresholds and response ownership; procurement planning defines make-or-buy, evaluation, contract and supplier controls.

WHY IT MATTERS

- Separate plans can conflict: a compressed schedule may require unavailable specialists; a low-cost supplier may not meet a quality threshold; a reporting cadence may arrive after the decision it supports. Integration checks reveal these trade-offs before execution.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Integrate Quality, Resources, Communications, Risk and Procurement Plans



In Practice — Integrate Quality, Resources, Communications, Risk and Procurement Plans

WORKED EXAMPLE

- The portal needs accessibility review before acceptance; the specialist is available only in week 9.
- The schedule reserves that window, procurement includes the review deliverable and the quality plan names the evidence.
- The risk plan records a trigger and contingency if the specialist becomes unavailable.

DECISION GUIDE

- **Use when**
 - Consolidating component plans and checking whether delivery assumptions agree across functions.
 - Preparing the project for execution, governance review or supplier engagement.
- **Avoid when**
 - Producing isolated documents that do not trace to deliverables, dates, costs and owners.
 - Asking AI to choose a supplier, staff member or response without approved criteria and human review.

Hands-On Labs — Project Initiation and Planning with GenAI

Lab 1

- Build the AI Working Agreement and Project Charter

Lab 2

- Build the Stakeholder Engagement and Communications Plan

Lab 3

- Build the Integrated Project Management Plan

Build the AI Working Agreement and Project Charter

LAB 1

You will open the synthetic BeaconWorks Customer Self-Service Portal brief, define a human-owned GenAI working agreement and turn the approved facts into a project charter. The exercise separates facts, assumptions, options and unknowns so polished language never becomes an unreviewed commitment.

You'll build

01-initiation/ai-working-agreement.md, project-charter.md and source-ledger.csv containing the approved project parameters, human review gates and traceable evidence.

1 Inspect evidence

2 Define AI controls

3 Draft the charter

4 Trace and approve

Tools: Text editor · approved AI assistant · beaconworks-project-brief.md · spreadsheet

Build the AI Working Agreement and Project Charter

Test it

The working agreement must contain all six headings and the six-part C-O-S-T-A-R pattern. The charter must contain all 13 requested sections, at least three measurable objectives, an explicit out-of-scope statement, a named document owner and approval owner, plus the status Ready for sponsor review. The source ledger must contain at least seven rows; every material figure and date in the charter must trace to a verified FACT row or be labelled ASSUMPTION or UNKNOWN.

Build the Stakeholder Engagement and Communications Plan

LAB 2

You will combine the approved charter with the supplied stakeholder inputs. GenAI will help structure and challenge the plan, while you make the final influence, interest, impact and engagement decisions from evidence. The resulting communication matrix starts with the action or decision each audience needs.

You'll build

02-planning/stakeholder-register.csv and stakeholder-communications-plan.md with eight stakeholder rows, current and desired engagement, owned actions, decision-focused communications and privacy safeguards.

1 Identify roles

2 Analyse evidence

3 Plan engagement

4 Design communications

Tools: Spreadsheet · text editor · approved AI assistant · stakeholder-inputs.csv · Lab 1 charter

Build the Stakeholder Engagement and Communications Plan

Test it

The register must contain exactly eight supplied Stakeholder_ID rows and no added personal data. Every row must have final influence, interest and impact scores; current and desired states; a strategy; an owned action; a review trigger; and evidence or UNKNOWN. The communications plan must contain at least five records and all 12 columns. The two audience drafts must preserve identical project facts and include different purpose-appropriate asks.

Build the Integrated Project Management Plan

LAB 3

You will use the charter, stakeholder plan and supplied work-package inputs to create a WBS-aligned baseline. You will calculate the network and budget yourself, then use GenAI to critique integration gaps across quality, resources, communications, risks and procurement.

You'll build

02-planning/wbs-and-baseline.csv, integrated-management-plan.md, initial-risk-register.csv and procurement-evaluation-plan.csv with a verified 28-working-day critical path and SGD 120,000 authorised budget.

1 Decompose scope

2 Sequence and estimate

3 Integrate controls

4 Verify the baseline

Tools: Spreadsheet · text editor · approved AI assistant · planning-inputs.csv · risk-issue-inputs.csv

Build the Integrated Project Management Plan

Test it

The baseline must contain ten contiguous work packages, total SGD 105,000 before contingency and show the zero-float path WP-01 -> WP-03 -> WP-04 -> WP-07 -> WP-08 -> WP-10 at 28 working days. Every row must have ES, EF, LS, LF and Total_Float calculated with the stated day-zero convention. The integrated plan must contain all 11 sections and link every work package to an owner, resource role, cost, quality or acceptance evidence and communication or control record. The risk register must contain five cause–event–effect rows with scored exposure, action, trigger, contingency and owner. Procurement criteria weights must total 100 and include written 1, 3 and 5 scoring anchors.

Recap — Project Initiation and Planning with GenAI

1

You can now: LO1: develop evidence-led project parameters and an authorised charter draft with GenAI

2

You can now: LO1: identify stakeholders and plan evidence-led engagement and communications

3

You can now: LO1: develop an integrated scope, schedule, cost, quality, resource, communication, risk and procurement plan

TOPIC 02

Project Execution and Delivery with GenAI

direct and manage work · knowledge and decisions · team and resources · quality · communications · stakeholder engagement · evidence-led status

Key Concepts — Project Execution and Delivery with GenAI

1

Work authorisation

Translate the plan into owned work packages with inputs, constraints and completion evidence.

2

Team coordination

Make responsibility, capacity, hand-offs, impediments and escalation routes visible.

3

Knowledge flow

Capture decisions, rationale, assumptions and lessons where the team can reuse them.

4

Quality at source

Prevent defects with clear criteria and peer checks instead of relying on late inspection.

5

Purposeful communication

Design each update around the decision or action the audience needs.

6

Evidence-led reporting

Separate facts, forecasts, decisions, risks and asks; never let narrative outrun the data.

Direct and Manage Project Work

WHAT

- Directing and managing work means authorising planned work packages, coordinating execution, producing deliverables, handling issues and implementing approved changes. The plan guides action, while actual results and work-performance data feed monitoring and decisions.

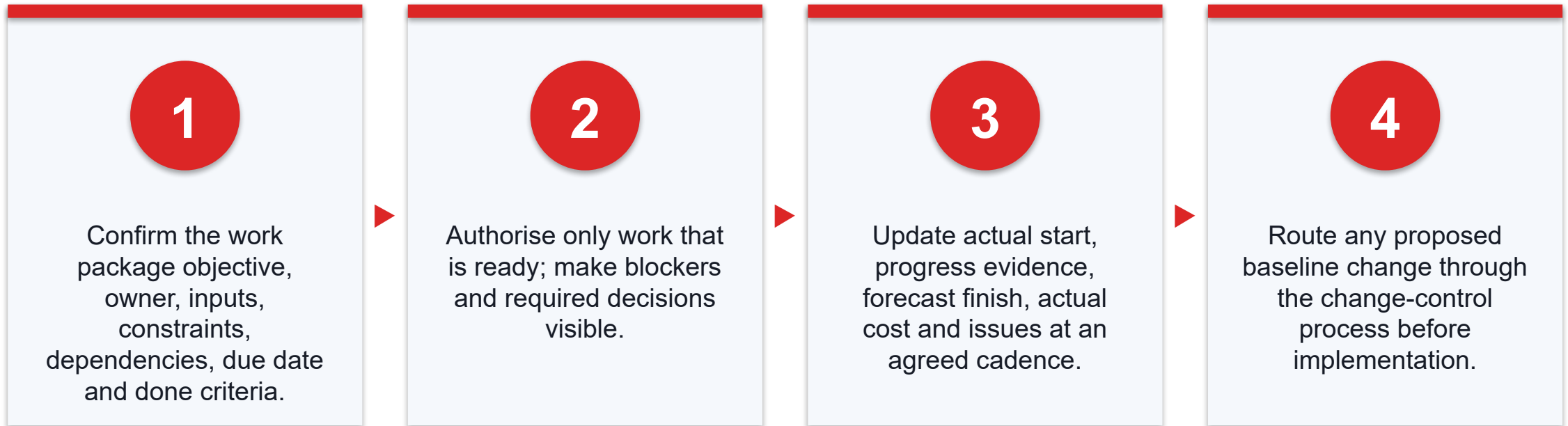
WHY IT MATTERS

- Execution fails when tasks are launched without clear outcomes, prerequisites or completion evidence. A work package brief creates a bounded commitment and lets AI assist with sequencing, checklists and communication without redefining the approved scope.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Direct and Manage Project Work



In Practice — Direct and Manage Project Work

WORKED EXAMPLE

- The configuration package starts only after content approval and environment access are confirmed.
- The owner records evidence links and a forecast, not an unsupported percent-complete guess.
- A new reporting feature is logged as a change request rather than inserted into active work.

DECISION GUIDE

- **Use when**
 - Converting an approved plan into short, owned units of delivery and coordinating dependencies.
 - Clarifying what can start, what is blocked and what evidence demonstrates completion.
- **Avoid when**
 - Starting work from an AI-generated list that has not been reconciled to the WBS and baseline.
 - Treating activity volume or polished status text as proof of deliverable progress.

Manage Project Knowledge and Decisions

WHAT

- Project knowledge includes explicit records—plans, decisions, designs and lessons—and tacit experience held by people. Knowledge management makes both usable through structured capture, context, ownership, retrieval and conversation. A decision log records the decision, options, evidence, rationale, owner, date and review trigger.

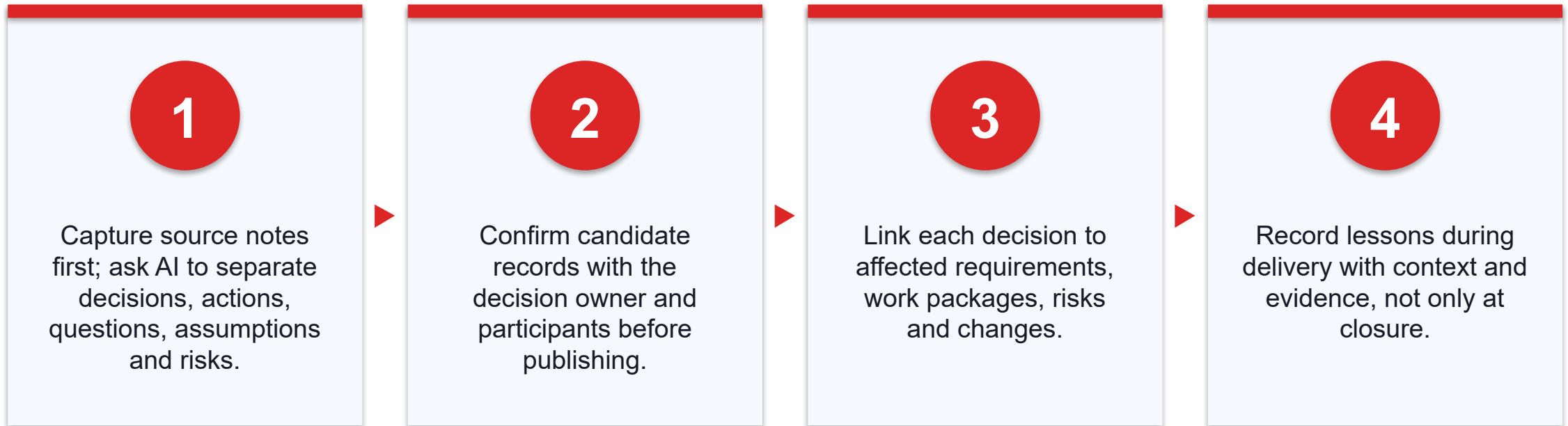
WHY IT MATTERS

- Meeting summaries without decisions and owners create the illusion of documentation. GenAI can extract candidate actions and themes, but source notes and participant confirmation are needed because a plausible summary can omit a condition or assign the wrong owner.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Manage Project Knowledge and Decisions



In Practice — Manage Project Knowledge and Decisions

WORKED EXAMPLE

- A design review chooses option B because it meets accessibility and lead-time constraints.
- The log retains rejected options, evidence, decision owner and a review trigger if the supplier date changes.
- The AI summary is corrected against the meeting notes before the action register is updated.

DECISION GUIDE

- **Use when**
 - Synthesising meetings, comparing options, onboarding team members or preserving rationale.
 - A project contains hand-offs, repeated decisions or knowledge that could be lost when people change.
- **Avoid when**
 - Recording confidential discussion in an unapproved service or treating an AI summary as the official record.
 - Capturing lessons as generic advice with no context, consequence or evidence.

Acquire Resources and Manage the Team

WHAT

- Resource management matches required capability and capacity to scheduled work, then supports people to deliver together. A responsibility view clarifies who is accountable, responsible, consulted and informed; a capacity view compares demand with realistic availability; a team working agreement defines collaboration, escalation and decision norms.

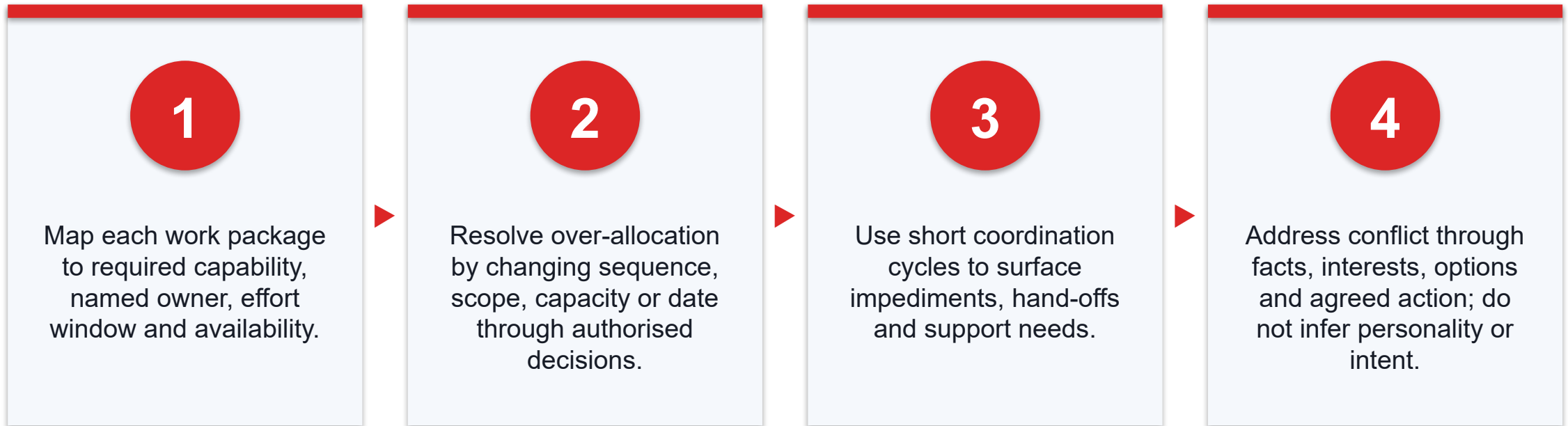
WHY IT MATTERS

- A plan can look feasible while over-allocating the same specialist or ignoring operational duties. GenAI can highlight conflicts and draft coaching questions, but staffing, performance and conflict decisions require context, fairness and accountable human judgement.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Acquire Resources and Manage the Team



In Practice — Acquire Resources and Manage the Team

WORKED EXAMPLE

- The same accessibility specialist is needed by content and test work in week 9.
- The team sequences the reviews and protects the critical-path activity rather than assuming parallel capacity.
- The resource plan records the change and the stakeholder update explains the trade-off.

DECISION GUIDE

- **Use when**
 - Planning or rebalancing capacity, clarifying responsibility and coordinating cross-functional work.
 - A delivery delay may be caused by demand, skill or dependency rather than individual effort.
- **Avoid when**
 - Using AI to rank people, infer sensitive characteristics or make employment decisions.
 - Treating availability as 100 percent of working time or ignoring operational responsibilities.

Manage Quality and Deliverable Acceptance

WHAT

- Managing quality turns the quality plan into preventive practices, peer review, process checks, defect learning and improvement. Quality control checks deliverable results against criteria; scope validation obtains formal acceptance from the authorised stakeholder. These are related but distinct: a deliverable may meet an internal check yet still need customer acceptance.

WHY IT MATTERS

- Late inspection makes defects expensive and encourages subjective debate. Clear criteria and evidence let the team build quality into the work. GenAI can create checklists and classify defects from approved records, but the criteria and acceptance decision remain human-owned.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Manage Quality and Deliverable Acceptance



In Practice — Manage Quality and Deliverable Acceptance

WORKED EXAMPLE

- The portal content must meet the agreed reading level, link integrity and accessibility checks.
- A peer review catches missing alternative text before the acceptance demonstration.
- The service director accepts the pilot after the evidence pack and two minor actions are recorded.

DECISION GUIDE

- **Use when**
 - Defining done, preventing rework, preparing a review or obtaining deliverable acceptance.
 - A quality dispute needs objective criteria, evidence and a named decision owner.
- **Avoid when**
 - Using an AI score as the sole evidence of quality or silently changing criteria after delivery.
 - Confusing completion of internal work with formal acceptance of the deliverable.

Manage Communications and Stakeholder Engagement

WHAT

- Communication management creates, distributes, stores and retrieves project information for a defined purpose. Stakeholder engagement uses that information plus dialogue, participation and relationship actions to support decisions and adoption. Effective communication is measured by understanding and action, not by the volume of messages sent.

WHY IT MATTERS

- The same status dump does not serve a sponsor, delivery team and affected user. GenAI is helpful for transforming one verified evidence set into audience-specific drafts, provided that facts, uncertainty, tone and calls to action are checked before release.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Manage Communications and Stakeholder Engagement



In Practice — Manage Communications and Stakeholder Engagement

WORKED EXAMPLE

- The team receives a detailed blocker-and-owner view; the sponsor receives trend, tolerance, decision and recommendation.
- Both versions use the same approved weekly data and preserve the same red status.
- The project manager removes unsupported certainty from the AI draft before distribution.

DECISION GUIDE

- **Use when**
 - Preparing status updates, decision briefs, change communications or adoption conversations.
 - Several audiences need different levels of detail from the same verified source.
- **Avoid when**
 - Using AI to manufacture positive sentiment, conceal bad news or impersonate a stakeholder.
 - Sending generated text without checking names, figures, commitments and confidentiality.

Build Evidence-Led Status and Forecast Reporting

WHAT

- Evidence-led reporting reconciles approved baselines, a common status date and verified work, cost, quality, resource, risk, issue and decision records before drafting a message. Earned value uses planned value (PV), earned value (EV) and actual cost (AC) at the same cut-off: schedule variance $SV = EV - PV$, cost variance $CV = EV - AC$, $SPI = EV \div PV$ and $CPI = EV \div AC$. Forecasts remain assumptions until their method, remaining-work estimate and owner are explicit.

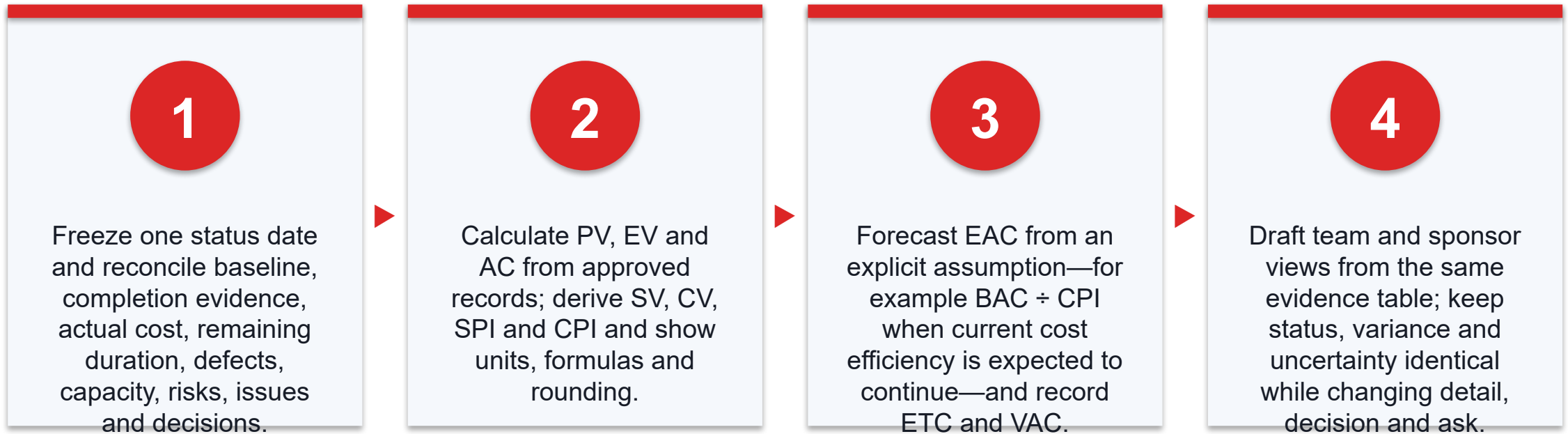
WHY IT MATTERS

- A polished status narrative can hide inconsistent dates, double-counted progress or unsupported optimism. A reconciled evidence table makes calculations reproducible and separates completed fact, current variance, forecast, risk, decision and ask. GenAI may transform the verified table for different audiences, but it must not calculate from mixed cut-offs or replace the project manager's judgement.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Build Evidence-Led Status and Forecast Reporting



In Practice — Build Evidence-Led Status and Forecast Reporting

WORKED EXAMPLE

- At week 8, PV is SGD 72,000, EV is SGD 60,000, AC is SGD 66,000 and BAC is SGD 105,000.
- SV is –SGD 12,000, CV is –SGD 6,000, SPI is 0.83 and CPI is 0.91; if CPI continues, EAC is SGD 115,500.
- The status remains red in both audience versions; the sponsor brief adds the decision needed while the team view adds owners and next-cycle evidence.

DECISION GUIDE

- **Use when**
 - Preparing a weekly status pack, variance analysis, estimate at completion, recovery recommendation or decision brief.
 - Different audiences need a consistent view of progress, cost, forecast, risk and action from one verified cut-off.
- **Avoid when**
 - Calculating earned value from unapproved scope, mixed status dates or progress percentages that lack completion evidence.
 - Using AI narrative confidence to hide an adverse variance, omit uncertainty or make a forecast appear authorised.

Hands-On Labs — Project Execution and Delivery with GenAI

Lab 4

- Turn the Baseline into an Executable Delivery System

Lab 5

- Manage Team Knowledge, Quality and Stakeholder Delivery

Lab 6

- Build the Evidence-Led Project Status Pack

Turn the Baseline into an Executable Delivery System

LAB 4

You will transform the frozen Baseline v1.0 into a delivery board without changing its scope or dates. The exercise distinguishes planned, ready, active, blocked and done work, then creates a detailed brief for portal configuration with prerequisites, acceptance evidence and escalation rules.

You'll build

03-delivery/delivery-board.csv, work-package-brief-WP-04.md and decision-log.csv that translate the baseline into controlled execution records.

1 Confirm readiness

2 Authorise work

3 Track evidence

4 Log decisions

Tools: Spreadsheet · text editor · approved AI assistant · Lab 3 baseline · delivery-events.md

Turn the Baseline into an Executable Delivery System

Test it

The board must contain the same ten work packages, baseline durations and costs as Baseline v1.0. Every Ready or Active row must have readiness evidence, owner, next action and objective progress evidence. The WP-04 brief must contain all 15 requested sections and no added scope. The decision log must contain E02 as Pending decision with at least two viable options, and E03 must be routed to a change request without editing the baseline.

Manage Team Knowledge, Quality and Stakeholder Delivery

LAB 5

You will process synthetic capacity, meeting and defect evidence without inferring people's intent. GenAI will extract candidate decisions and actions, while you confirm them against the source and update the resource, decision, quality and stakeholder records.

You'll build

03-delivery/resource-and-raci-plan.csv, quality-and-acceptance-log.csv, meeting-record-ai-draft.md, meeting-record.md, stakeholder-engagement-update.md and delivery-handoff-checklist.md with confirmed owners and evidence.

1 Reconcile capacity

2 Extract knowledge

3 Control quality

4 Update engagement

Tools: Spreadsheet · text editor · approved AI assistant · delivery-events.md · Lab 2–4 records

Manage Team Knowledge, Quality and Stakeholder Delivery

Test it

The resource plan must use one row per resource role and show these E04 results: Accessibility Specialist -2/YES, Integration Engineer 0/NO, Content Lead -2/YES and Support Lead +1/NO for Variance_Days/Conflict. The meeting record must classify at least one decision, action, question, assumption, risk and lesson, each with evidence and confirmation status. The quality log must keep verification result separate from acceptance status and include owners for all corrective actions. The engagement update must use observable evidence and the handoff checklist must be saved as 03-delivery/delivery-handoff-checklist.md and contain all seven required items.

Build the Evidence-Led Project Status Pack

LAB 6

You will reconcile the baseline, delivery board and supplied week-8 update. You will calculate schedule and cost indicators in the spreadsheet before asking GenAI to draft narrative. The final pack distinguishes fact, forecast, risk, decision and ask for both sponsor and team audiences.

You'll build

03-delivery/week-08-performance.csv and week-08-status-pack.md containing transparent calculations, source links, a sponsor decision brief and a delivery-team action view.

1 Set status date

2 Calculate variance

3 Forecast outcome

4 Tailor the message

Tools: Spreadsheet · text editor · approved AI assistant · weekly-update-inputs.md · Lab 3–5 records

Build the Evidence-Led Project Status Pack

Test it

The spreadsheet must show SV -SGD 12,000, CV -SGD 6,000, SPI 0.83, CPI 0.91, EAC SGD 115,500, ETC SGD 49,500 and VAC -SGD 10,500. The pack must show a 32-day forecast against the 28-day baseline, apply the stated status threshold consistently, contain source IDs for every material claim, preserve identical facts across sponsor and team views, and include a complete decision request plus nine-item next-cycle checklist.

Recap — Project Execution and Delivery with GenAI

1

You can now: LO2: direct and manage project work with owned work packages, dependencies and completion evidence

2

You can now: LO2: resolve delivery challenges through explicit capacity, knowledge, quality and engagement controls

3

You can now: LO2: communicate project delivery status and budget challenges from verified work-performance evidence

TOPIC 03

Manage Project Risks with GenAI

risk and issue response · procurement · monitor and control work · integrated change · scope, schedule, cost, quality, resources, communications, stakeholders and closure

Key Concepts — Manage Project Risks with GenAI

1

Risk response

Turn priority uncertainty into preventive action, triggers, contingencies, owners and residual exposure.

2

Integrated control

Compare actual and forecast performance with approved baselines and explain the cause and consequence.

3

Change control

Evaluate a proposed change across value, scope, schedule, cost, quality, resources, risk and contracts before deciding.

4

Forecast discipline

Use transparent formulas and assumptions; treat AI narrative as a draft around verified numbers.

5

Acceptance and transition

Close only after deliverables, obligations, handover, records and open actions have named owners.

6

Learning loop

Capture reusable lessons with context, evidence and a future action, then feed them into the next project.

Implement Risk Responses and Manage Issues

WHAT

- A risk is an uncertain event or condition that may affect objectives; an issue is a condition that has occurred and needs action. Response planning selects a strategy, owner, preventive action, trigger, contingency and fallback. Implementation executes those actions and checks whether exposure has changed or secondary risks have appeared.

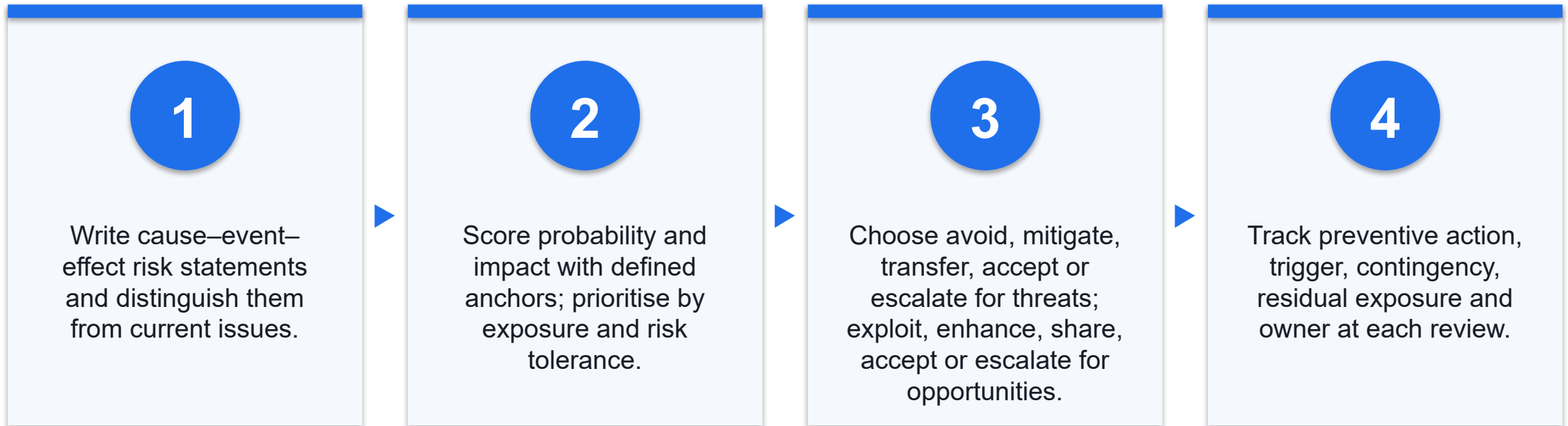
WHY IT MATTERS

- A colourful risk register does not reduce exposure. Actionable records connect priority uncertainty to funded, scheduled work and decision triggers. GenAI can challenge causes, consequences and response completeness, but probability, impact and ownership must be grounded in project evidence.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Implement Risk Responses and Manage Issues



In Practice — Implement Risk Responses and Manage Issues

WORKED EXAMPLE

- Cause: supplier API access may be delayed; event: test access is unavailable by week 8; effect: integration testing and launch move.
- Mitigation secures a sandbox date; trigger is no confirmed access by Friday; contingency uses a stub for non-production testing.
- When access misses the trigger, the record becomes an issue and the contingency starts.

DECISION GUIDE

- **Use when**
 - Preparing risk reviews, selecting response actions or converting a triggered risk into issue work.
 - The team needs a consistent way to compare threats, opportunities and response urgency.
- **Avoid when**
 - Inventing probabilities from no data or asking AI to own a risk.
 - Listing vague concerns with no cause, effect, trigger, action, date or responsible owner.

Conduct and Control Procurements

WHAT

- Procurement management selects an appropriate sourcing approach, defines requirements and evaluation criteria, obtains authorised offers, manages contract interfaces and monitors supplier performance. Project controls must include supplier deliverables, dependencies, costs, risks, changes, acceptance and obligations.

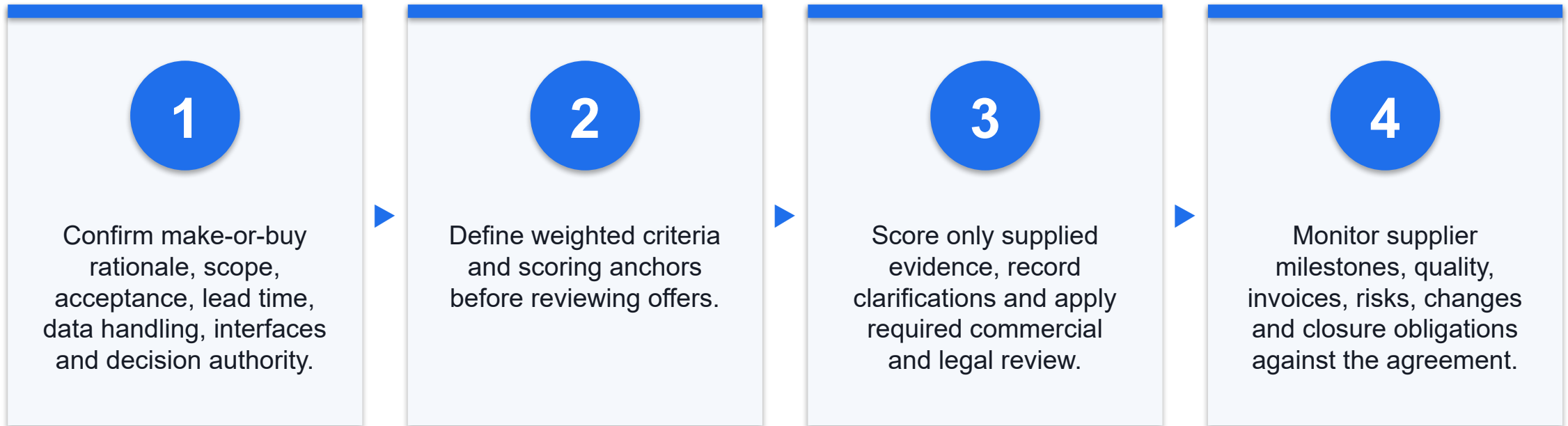
WHY IT MATTERS

- The cheapest offer is not necessarily the best value, and an AI-generated comparison can hide missing criteria or inconsistent evidence. A weighted decision matrix makes approved criteria and evidence visible while preserving accountable commercial judgement.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Conduct and Control Procurements



In Practice — Conduct and Control Procurements

WORKED EXAMPLE

- Three accessibility-review offers are compared on capability, approach, lead time, evidence and cost.
- AI extracts comparable fields, but missing evidence receives no invented score and procurement validates the source.
- The selected supplier's dates and acceptance evidence are integrated into the project baseline.

DECISION GUIDE

- **Use when**
 - Planning a purchase, comparing offers or monitoring external deliverables and obligations.
 - A supplier dependency affects the critical path, quality, risk or acceptance.
- **Avoid when**
 - Sharing confidential offers with an unapproved AI service or delegating the award decision to AI.
 - Changing supplier scope informally without authorised project and contract change records.

Monitor and Control Project Work

WHAT

- Monitoring and control compare actual and forecast results with the approved plan, analyse variance and trends, evaluate consequences and recommend action. Useful control integrates scope, schedule, cost, quality, resources, risks, communications and supplier information at a consistent status date.

WHY IT MATTERS

- Percent complete and traffic-light colours can obscure the basis of a forecast. Controls are credible when they use traceable source data, defined thresholds and transparent calculations. GenAI may explain verified results and find contradictions, but it should not calculate from ambiguous fields or rewrite a red condition as green.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Monitor and Control Project Work



In Practice — Monitor and Control Project Work

WORKED EXAMPLE

- At week 8: planned value is SGD 72,000, earned value SGD 60,000 and actual cost SGD 66,000.
- Schedule performance index = $60/72 = 0.83$; cost performance index = $60/66 = 0.91$.
- The project is behind plan and over cost for the work achieved; the report explains the integration-test dependency and recovery options.

DECISION GUIDE

- **Use when**
 - Preparing a status review, diagnosing variance or forecasting whether objectives remain achievable.
 - Several control dimensions must be reconciled before a decision.
- **Avoid when**
 - Using narrative confidence without source data, a status date or baseline reference.
 - Applying earned-value formulas when scope, schedule and cost are not integrated enough to support them.

Perform Integrated Change Control

WHAT

- Integrated change control records a proposed change, analyses its total impact, obtains a decision from the authorised body and updates affected baselines and records only after approval. It also communicates the decision and verifies implementation. Change control protects value and traceability; it is not a promise to reject change.

WHY IT MATTERS

- A small feature can alter design, testing, schedule, cost, support, supplier work and risk. GenAI is useful for building an impact checklist and comparing options, but its proposal must be reconciled with the WBS, schedule, estimates, contracts and stakeholder evidence.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Perform Integrated Change Control



In Practice — Perform Integrated Change Control

WORKED EXAMPLE

- A bilingual FAQ is requested in week 8. The analysis adds content, translation, accessibility checks and support training.
- Option A adds SGD 9,000 and eight working days; option B pilots ten priority articles within tolerance.
- The sponsor approves option B; scope, schedule, cost, risk and stakeholder records are updated together.

DECISION GUIDE

- **Use when**
 - A requirement, deliverable, baseline, contract or control threshold may change.
 - The team needs a traceable decision and coordinated update across project records.
- **Avoid when**
 - Implementing the request before authorisation or updating only the schedule while leaving scope and cost unchanged.
 - Treating an AI recommendation as approval or omitting rejected options and rationale.

Control Scope, Schedule, Costs, Quality and Resources

WHAT

- Control maintains alignment between approved baselines and actual work. Scope control prevents unapproved expansion and preserves traceability; schedule control updates logic and forecasts; cost control tracks actuals and estimates; quality control evaluates results; resource control compares required and available capability. Corrective action is integrated because one dimension can shift another.

WHY IT MATTERS

- Local optimisation can worsen the whole project—for example, adding people may increase cost and coordination load without recovering a late dependency. A control decision should state evidence, options, trade-offs and the authorised path.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Control Scope, Schedule, Costs, Quality and Resources



In Practice — Control Scope, Schedule, Costs, Quality and Resources

WORKED EXAMPLE

- Integration testing is late because supplier access moved, not because test effort increased.
- The team protects acceptance criteria, resequences training preparation and uses the authorised stub contingency.
- The forecast, cost impact, residual risk and stakeholder message are updated from the same decision.

DECISION GUIDE

- **Use when**
 - Correcting performance, reforecasting, protecting scope or deciding how to use limited capacity.
 - A variance crosses tolerance or affects another baseline dimension.
- **Avoid when**
 - Changing a baseline to make current performance look acceptable.
 - Recommending action from one metric without analysing consequences for quality, risk, people and value.

Monitor Communications, Stakeholders and Risks

WHAT

- Monitoring checks whether communication is reaching the right people in time, whether stakeholder engagement is moving toward the desired state and whether risk exposure and response effectiveness have changed. The team updates plans from evidence such as decisions delayed, questions repeated, actions missed, sentiment expressed directly and triggers observed.

WHY IT MATTERS

- A communication can be sent yet fail to produce understanding or action. Stakeholder and risk conditions also change as delivery progresses. GenAI can cluster feedback and surface contradictions from approved material, but it must not infer hidden motives or treat sentiment labels as facts.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Monitor Communications, Stakeholders and Risks



In Practice — Monitor Communications, Stakeholders and Risks

WORKED EXAMPLE

- Support agents repeat the same launch-readiness question, showing that the FAQ update did not create understanding.
- The project replaces a one-way email with a live walkthrough and captures unresolved questions.
- The adoption risk score and communication plan are updated after the session.

DECISION GUIDE

- **Use when**
 - Checking the effectiveness of a status cycle, change communication, adoption activity or risk response.
 - A stakeholder delay or misunderstanding threatens a milestone or acceptance.
- **Avoid when**
 - Inferring private attitudes from sparse text or storing unnecessary personal commentary.
 - Closing a risk because an action was completed without checking residual exposure.

Close the Project or Phase

WHAT

- Closure confirms that approved work is complete or formally transferred, acceptance is recorded, contracts and obligations are settled, operations can support the outcome, records are archived, remaining actions have owners and lessons are reusable. Closure may also document an authorised early termination and its consequences.

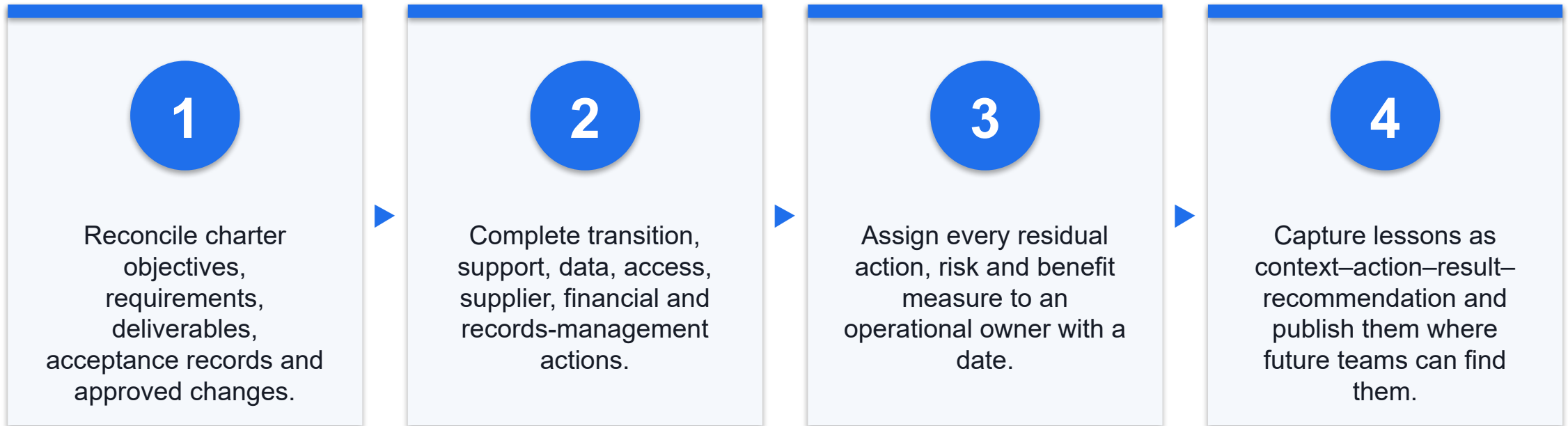
WHY IT MATTERS

- A launch date is not the same as closure. Unaccepted deliverables, open supplier obligations, unsupported operations or undocumented decisions create hidden work and risk. GenAI can help compile a closeout pack from verified records, but each acceptance, financial and contractual fact needs an authoritative source.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Close the Project or Phase



In Practice — Close the Project or Phase

WORKED EXAMPLE

- The pilot is accepted with two minor actions transferred to the service owner and dated.
- Supplier acceptance, final invoice, access removal, support runbook and benefits review are recorded.
- A lesson about early content approval includes evidence, consequence and a reusable planning action.

DECISION GUIDE

- **Use when**
 - Completing a project or phase, transitioning ownership or documenting an authorised stop.
 - Preparing an evidence pack for sponsor confirmation and future project reuse.
- **Avoid when**
 - Declaring closure because the team stopped working or because an AI summary says all items are complete.
 - Leaving open actions, risks, benefits or support obligations without named operational owners.

Hands-On Labs — Manage Project Risks with GenAI

Lab 7

- Build the Risk, Issue and Procurement Response Cockpit

Lab 8

- Run Integrated Change Control and Baseline Impact Analysis

Lab 9

- Close the Project and Capture Reusable Lessons

Build the Risk, Issue and Procurement Response Cockpit

LAB 7

You will refresh the initial risk register at the week-8 status date, distinguish triggered risks from current issues and compare three synthetic accessibility-review offers against criteria written in Lab 3. GenAI may structure evidence and challenge gaps, but final scores, owners and recommendation remain human-owned.

You'll build

04-control/risk-issue-cockpit.csv, procurement-comparison.csv and response-brief.md with current exposure, triggered actions, weighted supplier evidence and integrated project impacts.

1 Refresh exposure

2 Trigger responses

3 Score offers

4 Integrate actions

Tools: Spreadsheet · text editor · approved AI assistant · risk-issue-inputs.csv · vendor-bids.csv

Build the Risk, Issue and Procurement Response Cockpit

Test it

All five initial risks must appear in the cockpit, with week-8 status, trigger result, current or residual exposure, owned action and review point. Conditions that have occurred must have Issue_IDs and must not remain worded only as future uncertainty. Procurement weights must remain unchanged and totals must be Vendor A 3.90, Vendor B 4.10 and Vendor C 3.70. The response brief must recommend—not award—Vendor B, integrate eight impact dimensions and contain the stated monitoring cadence.

Run Integrated Change Control and Baseline Impact Analysis

LAB 8

You will evaluate a request for bilingual pilot content at week 8. The exercise prevents a useful feature from bypassing scope, schedule, cost, quality, resource, risk, supplier, communication and operational review. You will model options, record the synthetic sponsor decision and publish a traceable Baseline v1.1.

You'll build

04-control/change-log.csv, change-impact-analysis-CR-004.md, baseline-v1.1.csv and change-implementation-checklist.md with an approved, fully traced incremental change.

1 Log the request

2 Analyse options

3 Record the decision

4 Update controlled records

Tools: Spreadsheet · text editor · approved AI assistant · change-request-CR-004.md · Labs 3, 6 and 7 records

Run Integrated Change Control and Baseline Impact Analysis

Test it

CR-004 must have a complete request record, four analysed options and at least 12 impact dimensions. The analysis must show full scope at +SGD 9,000/+8 days and the approved pilot at +SGD 3,500/+3 days. Baseline v1.1 must show SGD 108,500 work budget, SGD 11,500 remaining contingency, unchanged SGD 120,000 authorised total and 31 working days. The implementation checklist must contain all 12 controlled-record updates with owner, evidence and verifier. Baseline v1.1 must add WP-09A and make WP-10 depend on it, while week8-performance.csv, delivery-board.csv and the week-8 status pack remain frozen and unchanged.

Close the Project and Capture Reusable Lessons

LAB 9

You will use the final synthetic evidence to determine whether the BeaconWorks pilot is ready to close. GenAI will help compile and challenge the pack, while you trace each statement to final records, transfer remaining actions and benefits, and record the synthetic sponsor's closure decision.

You'll build

05-closure/closeout-checklist.csv, project-closeout-report.md, transition-and-benefits-plan.md and lessons-register.csv with formal acceptance, residual ownership and reusable context.

1 Reconcile completion

2 Confirm acceptance

3 Transition ownership

4 Publish lessons

Tools: Spreadsheet · text editor · approved AI assistant · closure-inputs.md · complete Labs 1–8 portfolio

Close the Project and Capture Reusable Lessons

Test it

The checklist must cover all 14 named categories, trace each completion or acceptance to evidence and show no unresolved Closure_Blocker. Every residual action, risk and benefit must have one owner and date in the transition plan. The lessons register must contain at least six evidence-backed C-A-R-R records with reuse triggers. The closeout report must contain all 14 named report sections plus the completed Claim verification table, reconcile final schedule and cost with the supplied evidence, record the sponsor closure decision and point to the final records location.

Recap — Manage Project Risks with GenAI

1

You can now: LO3: anticipate and implement risk, issue and supplier responses from verified evidence

2

You can now: LO3: evaluate and decide a project change across all affected controls with GenAI support

3

You can now: LO4: validate scope, obtain acceptance, transition ownership and close with traceable records



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Initiate & Plan

Develop project parameters and an integrated project plan with GenAI according to business requirements.

2

Execute & Deliver

Resolve project delivery challenges by using GenAI to support people, knowledge, quality, communications, resources and budgets.

3

Respond & Change

Anticipate, monitor and respond to project risks and changes with GenAI while keeping decisions evidence-led and human-owned.

4

Control & Close

Control project performance, obtain acceptance and close a project or phase with traceable records and reusable lessons.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Keep your downloaded C013 course files and use the lab checkpoints to restart safely.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

THANK YOU FOR ATTENDING

Thank You!

You now have the practical skills from every lab in this course — keep applying them to your own work.